

Math 252, Third Examination Review

Directions: Find a solution to each problem in the space provided. You may use the back of the page if necessary. Show all work in order to receive credit. No calculators or electronic devices are permitted on this exam.

1. Let $f(x) = \ln(1 + 2x)$.

(a) Using derivatives, determine the fourth degree Taylor polynomial around $x = 0$ for f .

(b) Using part (a) determine the general term, and use it to evaluate $f^{(10)}(0)$.

2. Let $f(x) = e^{-x^2}$.

(a) Using derivatives, determine the third degree Taylor polynomial around $x = 0$ for f .

(b) Using part (a) determine the general term, and use it to evaluate $f^{(7)}(0)$.

3. (a) Determine if the following series is convergent, and find the sum if it is: $\sum_{n=0}^{\infty} \frac{2^n}{3^{2n}}$.

(b) Determine if the following series is convergent, and find the sum if it is: $\sum_{n=0}^{\infty} \frac{2^{n-1}}{3^{n+1}}$.

(c) Determine the sum of the following telescoping series: $\sum_{n=1}^{\infty} \left(\frac{1}{n} - \frac{1}{n+2} \right)$

(d) Determine the sum of the following telescoping series: $\sum_{n=2}^{\infty} \left(\frac{1}{n+1} - \frac{1}{n+3} \right)$

4. Determine if the following series converge or diverge, giving reasons. Show all work and justify all responses to receive credit.

(a) $\sum_{n=1}^{\infty} \frac{2n}{n^3 + 1}$

(b) $\sum_{n=1}^{\infty} \frac{n^2}{e^n}$.

(c) $\sum_{n=1}^{\infty} \frac{3n^2 + 1}{2n^2 + 4n + 1}$.

(d) $\sum_{n=2}^{\infty} \frac{1}{n(\ln n)^2}$

(e) $\sum_{n=1}^{\infty} \frac{2n^3 - 1}{n^3 - 1}$

5. Consider the series $\sum_{n=1}^{\infty} (-1)^n \frac{1}{\sqrt{n+5}}$. Determine if the series converges absolutely, converges conditionally, or diverges. Give reasons.

(e) A p -series

$$\sum_{n=1}^{\infty} \frac{1}{n^p}$$

converges if

and diverges if

_____.

(f) If

$$\sum_{n=1}^{\infty} |a_n|$$

converges, then

$$\sum_{n=1}^{\infty} a_n$$

converges

_____.

(g) If

$$\sum_{n=1}^{\infty} a_n$$

converges, but

$$\sum_{n=1}^{\infty} |a_n|$$

does not, then the series converges

_____.

(h) An alternating series

$$\sum (-1)^n b_n$$

converges provided that b_n is eventually

and

$$\lim_{n \rightarrow \infty} b_n = \text{_____}.$$

Tests for Convergence

Test	Conclusion
Divergence Test	If $\lim_{n \rightarrow \infty} a_n \neq 0$ (or DNE), then $\sum a_n$ diverges.
Geometric Series	$\sum ar^n$ converges for $ r < 1$, diverges for $ r \geq 1$.
p -Series	$\sum \frac{1}{n^p}$ converges if $p > 1$; diverges if $p \leq 1$.
Telescoping Series	Write the partial sums. If the remaining terms approach a finite limit, the series converges.
Integral Test	If $a_n = f(n)$ where f is positive, continuous, and decreasing, then $\sum a_n$ and $\int_0^\infty f(x) dx$ either both converge or both diverge.
Direct Comparison	If $0 \leq a_n \leq b_n$, then $\sum b_n$ convergent implies $\sum a_n$ convergent, and $\sum a_n$ divergent implies $\sum b_n$ divergent.
Limit Comparison	If $\lim_{n \rightarrow \infty} \frac{a_n}{b_n} = L$, $0 < L < \infty$, then $\sum a_n$ and $\sum b_n$ have the same behavior.
Alternating Series	If b_n is decreasing and $\lim b_n = 0$, then $\sum (-1)^n b_n$ converges.
Absolute Convergence	If $\sum a_n $ converges, then $\sum a_n$ converges.
Ratio Test	Let $L = \lim \left \frac{a_{n+1}}{a_n} \right$. $L < 1$: Converges absolutely $L > 1$: Diverges $L = 1$: Inconclusive
Root Test	Let $L = \lim \sqrt[n]{ a_n }$. $L < 1$: Converges absolutely $L > 1$: Diverges $L = 1$: Inconclusive